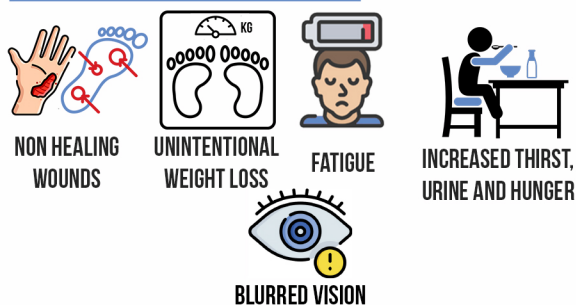




INDICATIONS FOR SCREENING

- **Present profile:** Pre-diabetes*, High Waist Circumference (≥ 80 cm in women, ≥ 90 cm in men), BMI (kg/m^2) > 23 , Sedentary lifestyle, Hypertension, High lipids, age ≥ 25 , Every pregnant women
- **Past history:** Gestational Diabetes, Stroke, Transient Ischemic Attack, Coronary Artery Disease
- **Personal history:** Smoking
- **Family history:** First degree relative with Diabetes Mellitus

SYMPTOMS OF HYPERGLYCEMIA (USUALLY ASYMPTOMATIC)



* DIAGNOSIS OF PRE-DIABETES

- HbA1c: 5.7 to 6.4% OR **at least 2 sugar level of**
- Fasting Plasma Glucose: 110-125mg/dL OR
- Post-Prandial Plasma Glucose: 140 -199mg/dL

Pre-diabetes is a stage where blood sugar is not high enough to label as diabetes, but are above normal values. **Progression of Type 2 diabetes can be prevented by healthy life style and weight loss.**

DIAGNOSIS OF TYPE 2 DIABETES MELLITUS

If any two of the following are present:

- HbA1c $\geq 6.5\%$ OR **at least 2 sugar level of**
- Fasting Plasma Glucose ≥ 126 mg/dl
- Post-Prandial Plasma Glucose ≥ 200 mg/dl
- Random Plasma Glucose > 200 +Classical Symptoms

Note: Fasting Blood Sugar (FBS) should be done after 8-12 hours of fasting
Post-Prandial Blood Sugar (PPBS) should be done after 2 hours of meal

LIST OF DRUGS

Class: Biguanides Drug: Metformin * Do not use if GFR < 30 ml/min	Side effects Diarrhea, Nausea, Flatulence, Vitamin B12 deficiency
Class: Sulphonylureas Drug: Gliclazide* , Glipizide* , Glimepiride**	Side effects Hypoglycemia, weight gain
Class: DPP4 inhibitors Drug: Linagliptin** , Sitagliptin**	Side effects Nasopharyngitis, Joint pain, Nausea, Pancreatitis
Class: SGLT2 inhibitors Drug: Empagliflozin** , Dapagliflozin	Side effects Genitourinary infection, Vulval candidiasis

*National list of essential medicines, 6th revision, MOHP Nepal

**Health insurance board benefit package, January 2024

SYMPTOMS OF HYPOGLYCEMIA



“ EACH PATIENT SHOULD BE AWARE OF HYPOGLYCEMIC SYMPTOMS AND INGEST 3 - 4 LUMPS OF SUGAR AS SOON AS THEY DEVELOP IT ”

- Give 3-4 teaspoons of sugar (if patient is able to eat/drink)
- Repeat if continued hypoglycaemia after 15 minutes for 3 consecutive times
- Give intravenous administration 25% 100 ml dextrose followed by monitoring of blood glucose every 15 minutes (if patient is not able to eat/drink)

OUR COLLABORATORS



Ref. PEN Manual (MoHP), 2075



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TYPE 2 DIABETES MELLITUS IDENTIFICATION AND MANAGEMENT FOR PRIMARY CARE PROVIDERS

USE THIS GUIDE TO

- Identify and approach patients with Type 2 Diabetes Mellitus
- Screen for diabetes among patients at risk
- Manage and refer patients with complications

Diabetes mellitus is a condition where there is high sugar levels in the blood. It is a common cause of kidney damage, heart disease, blindness, chronic non healing wounds, etc.

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INVESTIGATIONS

FBS, PPBS, HbA1c, URINARY ALBUMIN, SERUM CREATININE, URINARY KETONES*, LIPID PROFILE* (*If available)

TARGET BLOOD PRESSURE & SUGAR

HbA1c : 6.5 - 7%

FBS : 80-130 mg/dl

PPBS : 130-180 mg/dl

Target Blood Pressure < 130/80

Note: Good glycemic control can help minimize complications related to diabetes

(Less stringent control in patients with multiple co-morbidities, cognitive impairment, frailty, advanced complications or short life expectancy)



NON-PHARMACOLOGICAL ADVICE



Regular Physical Activity



Reduce Stress



Low Salt Consumption



Quit Alcohol & Tobacco



Consume Healthy Diet



Weight Reduction



Adequate Sleep

Note: All women with diabetes must be given pre-conception counselling and pregnant women with diabetes should be referred to higher center



PHARMACOLOGICAL TREATMENT

Start with Metformin 500 mg po BD

2-4 WEEKS

If the blood sugar is not controlled then

Increase to Metformin 1000mg BD

3 MONTHS

If the blood sugar is not controlled then Add

Tab. Glimepiride 1mg OD / Gliclazide 40mg OD

3 MONTHS

If the blood sugar is not controlled then

increase Glimepiride to 2 mg (maximum dose of 4 mg per day in divided doses) or Gliclazide 80 mg (maximum dose of 160mg OD)

3 MONTHS

If the blood sugar is not controlled then

Add Sitagliptin 100 mg OD / Linagliptin 5mg OD

3 MONTHS

If the blood sugar is not controlled then Refer

Note: SGLT2 inhibitors (Empagliflozin): Benefit in overweight, CKD, Heart disease (**if available)

Note: Give tablet **Atorvastatin 10-20mg** to all patients aged > 40 years and less than 75 years of age. Take all measure to help quit smoking.



INDICATIONS FOR INSULIN (NEEDS REFERRAL)

1. Uncontrolled DM after 3 oral drugs in a compliant patient
2. May be considered if:
 - Evidence of ongoing catabolism (weight loss)
 - Presence of symptoms of hyperglycemia (polydipsia, polyphagia, polyuria)
 - HbA1c > 10%
 - Blood glucose level > 300mg/dl



CRITERIA FOR REFERRAL

- Uncontrolled diabetes (HbA1c ≥ 10, blood sugar more than 300 mg/dl) despite treatment
- Presence of heart disease, stroke
- Unconsciousness
(Refer after giving intravenous 25% dextrose solution followed by monitoring of blood glucose every 15 minutes if possible)
- Severe infection
- Acute painful limb
- Diabetic kidney disease
(Estimated glomerular filtration rate (eGFR) (use eGFR calculator) of <60 mL/min per 1.73 m² on at least two occasions, 1 to 3 months apart, and/or Presence of albuminuria in at least two urine samples, 1 to 3 months apart)
- Decreased vision, neuropathy or diabetic foot
- Diabetes with pregnancy

**ALSO REFER TO
KHDC CVD RISK ASSESSMENT LEAFLET**



SCREENING FOR COMPLICATIONS

EVERY YEAR LOOK FOR ALBUMIN IN URINE

EVERY YEAR PERFORM RETINAL EXAMINATION, AND PALPATE PERIPHERAL PULSES, PROVIDE PROPER DENTAL CARE

EVERY VISIT PERFORM SKIN INSPECTION OF THE FOOT, ASSESMENT OF BP, COUNSELING

CONTACT US

For any clinical queries / If anything in the document is difficult to understand



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Disclaimer: This leaflet is currently being piloted and may contain inconsistencies. Please help us improve it by reporting any issues to KHDC Nepal.